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Computational Marine Ecological Modeling
Report 1: NPZDO model

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1 State variables & Processes

The NPZDO model in a water column for a close system includes 5 state variables namely, nutrient (N), phytoplankton (P), zooplankton (Z), detritus (D) and oxygen (O). The nutrient in the model is currently nitrogen hence the model's currency is N (i.e. mmolN) for nitrogen, except for oxygen where the unit is mmolO_2. The model forcings include only light (IO) with unit of μ mol photons. The model's state variables and processes are shown in Figure 1. The model includes both biochemical processes as well as physical processes (i.e. advection and diffusion) in the water column.

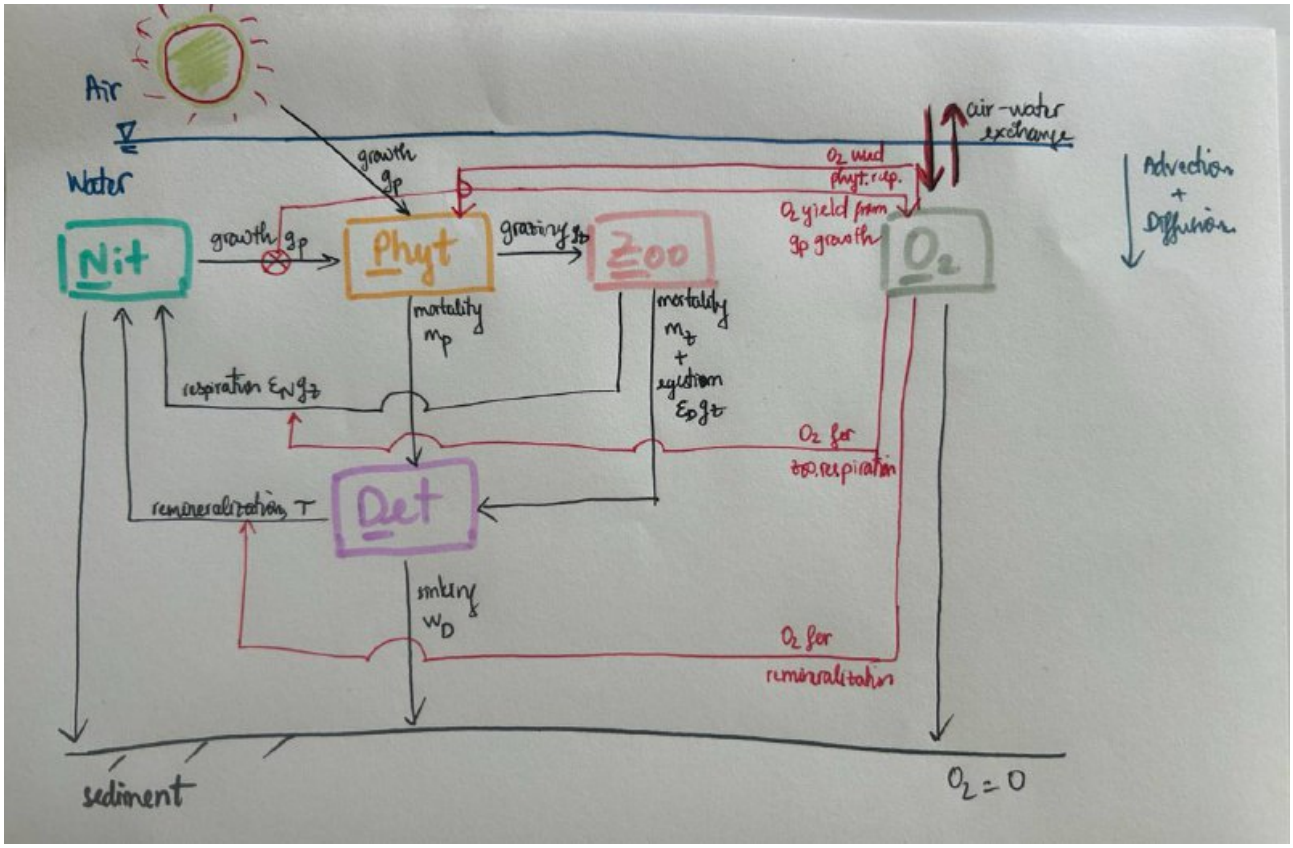


Figure 1: State variables in boxes and relevant processes included in the model.

2 Governing equations

The governing equations for the 5 state variables are below. Note that the terms in bracket refers to diffusion and advection terms for each state variables.

$$\frac{dN}{dt} = -g_P P + \epsilon_N g_Z Z + \tau D + \left(\kappa_{(z,t)} \frac{dN}{dz}\right) \tag{1}$$

$$\frac{dP}{dt} = g_P P - g_Z Z - m_P P + \left(\kappa_{(z,t)} \frac{dP}{dz}\right) \tag{2}$$

$$\frac{dZ}{dt} = (1 - \epsilon_N - \epsilon_D) g_Z Z - m_Z Z + \left(\kappa_{(z,t)} \frac{dZ}{dz}\right) \tag{3}$$

$$\frac{dD}{dt} = m_P P + m_Z Z + \epsilon_D g_Z Z - \tau D + \left(\kappa_{(z,t)} \frac{dD}{dz}\right) \tag{4}$$

$$\frac{dO}{dt} = y_P g_P P - y_N \epsilon_N g_Z Z - y_N \tau D + \left(\kappa_{(z,t)} \frac{dO}{dz}\right) \tag{5}$$

Phytoplankton growth rate (g_{P}) is dependent on nutrient and light availability using half saturation constant k_{N} and k_{L} for nutrient and light respectively.

$$g_{\mathrm{P}} = g_{\mathrm{P,max}} \min\left\{\frac{\mathrm{L}}{\mathrm{L}+k_{\mathrm{L}}}, \frac{\mathrm{N}}{\mathrm{N}+k_{\mathrm{N}}}\right\} \tag{6}$$

Zooplankton consumption rate (g_{Z}) is dependent on phytoplankton concentration using half-saturation constant k_{Z}, and oxygen concentration using a sigmoid function of prescribed $\mathrm{O}_{2,\mathrm{crit}}$.

$$g_{\mathrm{Z}} = g_{\mathrm{Z,max}} \frac{\mathrm{P}}{\mathrm{P}+k_{\mathrm{Z}}} \frac{1}{1+\exp(\mathrm{O}_{2,\mathrm{crit}}-\mathrm{O})} \tag{7}$$

The terms used in the equations are largely the same as described in Figure 1, except for processes related to oxygen where:

- $y_{\mathrm{P}} g_{\mathrm{P}} \mathrm{P}$ = oxygen yield from phytoplankton growth (or production),
- $y_{\mathrm{N}} \epsilon_{\mathrm{N}} g_{\mathrm{Z}} \mathrm{Z}$ = oxygen consumption from zooplankton respiration,
- $y_{\mathrm{N}} \tau \mathrm{D}$ = oxygen consumption for remineralization.

Surface light in the model can be constant or seasonally varying. The seasonal variation of surface light is described below:

$$I_0(t) = 0.5\left[1+\sin\left(\frac{2\pi t}{365}\right)\right](I_{0,\mathrm{max}} - I_{0,\mathrm{min}}) + I_{0,\mathrm{min}} \tag{8}$$

where $I_{0,\mathrm{max}}$ is the summer peak surface light and $I_{0,\mathrm{min}}$ is the winter minimum surface light.

Furthermore, light is reduced with depth based on light attenuation (Beer-Lamber law, k_{w}) as well as self-shading by phytoplankton, k_{p}. Note that the z notion is depth below the surface (positive downward).

$$I(z,t) = I_0(t)\,\exp\left(-k_{\mathrm{W}} z \;-\; \int_0^z \mathrm{P}(z)\, k_{\mathrm{P}}\, dz\right) \tag{9}$$

The vertical mixing coefficient $\kappa_{(\mathrm{z,t})}$ can be constant, throughout time (t) and space (z) but can also vary throughout time and space. The equation to describe the vertical variation is described below.

$$\kappa(z) = 0.5\left(1-\tanh\left(\frac{z-z_{\mathrm{MixSeason}}}{\zeta_{\mathrm{mix}}}\right)\right)(\kappa_{\mathrm{top}} - \kappa_{\mathrm{bottom}}) \;+\; \kappa_{\mathrm{bottom}} \tag{10}$$

The seasonal varying mixing coefficient can be described by varying the thickness of the $z_{\mathrm{MixSeason}}$ where the mixed layer depth is varying based on season. In reality, $z_{\mathrm{MixSeason}}$ should be a function of additional forcings such as air temperature (i.e. related to water temperature) and salinity as these forcings drive the density of different layers in the water column.

$$z_{\mathrm{MixSeason}}(t) = \frac{1}{2}\left(1-\sin\left(\frac{2\pi\left(t-t_{\mathrm{MaxSpring}}\right)}{365\text{ days}}\right)^{\zeta_{\mathrm{MaxSteep}}}\right)(z_{\mathrm{MixWinter}} - z_{\mathrm{Mix}}) \;+\; z_{\mathrm{Mix}} \tag{11}$$

where:

- $z_{\mathrm{MixSeason}}(t)$ = seasonally varying mixing depth,
- z_{Mix} = reference mixing depth,
- $z_{\mathrm{MixWinter}}$ = maximum winter mixing depth (deep mixing),
- $t_{\mathrm{MaxSpring}}$ = time of max spring bloom,
- κ_{top} = diffusivity in upper layer,
- κ_{bottom} = diffusivity value in the deeper layer,

- ζ_{mix} = controls how sharp the transition is between mixed (upper) and unmixed layers. Smaller value indicates sharper thermocline and higher value indicates smoother gradient,
- ζ_{MaxSteep} = controls how steep the seasonal mixing depth transitions between summer and winter. Higher value indicate more abrupt seasonal transitions.

3 Parameters used

Table 1 describes the parameters used in the model.

Table 1: Model parameters.

Name	Symbol	Value	Unit
Water column			
Water column depth	z_{bottom}	20	m
Number of discretized cells	n	100	cells
Upper layer diffusivity	κ_{top}	10	m^2/day
Deeper layer diffusivity	κ_{bottom}	5	m^2/day
Reference mixing depth	z_{Mix}	10	m
Winter mixing depth	$z_{\text{MixWinter}}$	20	m
Time of max spring bloom	$t_{\text{MaxSpring}}$	75	day
Transition thickness	ζ_{mix}	2	m
Pycnocline slope exponent	ζ_{MaxSteep}	4	m
Settling velocity	w	0.1	day^{-1}
Light			
Max surface light (in summer)	$I_{0,\text{max}}$	1400	μmol ph/m^2/sec
Min surface light (in winter)	$I_{0,\text{min}}$	0	μmol ph/m^2/sec
Average surface light (in summer)	I_0	100	μmol ph/m^2/sec
Light attenuation (water)	k_{w}	0.15	m^{-1}
Light attenuation (phytoplankton)	k_{P}	0.05	m^2/mmolN
Phytoplankton			
Max phytoplankton growth rate	$g_{\text{P,max}}$	2	day^{-1}
Light half-saturation constant	k_{L}	20	μmol ph/m^2/sec
Nutrient half-saturation constant	k_{N}	0.3	mmolN/m^3
Phytoplankton mortality rate	m_{P}	0.07	day^{-1}
Zooplankton			
Max zooplankton grazing rate	$g_{\text{Z,max}}$	1	day^{-1}
Phytoplankton half-saturation constant	k_{Z}	1	mmolN/m^3
Zooplankton respiration rate	ϵ_{N}	0.3	-
Zooplankton excretion rate	ϵ_{D}	0.3	-
Zooplankton mortality rate	m_{Z}	0.2	day^{-1}
Detritus			
Detritus re-mineralization rate	τ	0.1	day^{-1}
Oxygen			
O_2 yield (phytoplankton growth)	y_{P}	9	mmolO_2/mmolN
O_2 consumed (remineralization)	y_{N}	6.625	mmolO_2/mmolN
Surface gas exchange coefficient	$k_{O_2,\text{surf}}$	10	m/day
Bottom gas exchange coefficient	$k_{O_2,\text{bottom}}$	1	m/day
Atmospheric O_2 saturation	$O_{2,\text{eq}}$	260	mmolO_2/m^3
Critical O_2	$O_{2,\text{crit}}$	100	mmolO_2/m^3
Bottom O_2	$O_{2,\text{sed}}$	0	mmolO_2/m^3

4 Numerical schemes & boundary conditions

The model describes 1D (vertical) water column of depth z_{bottom} divided into n cells of equal thickness $\Delta z = z_{\text{bottom}}/n$. The vertical discretization is described in Figure 2.

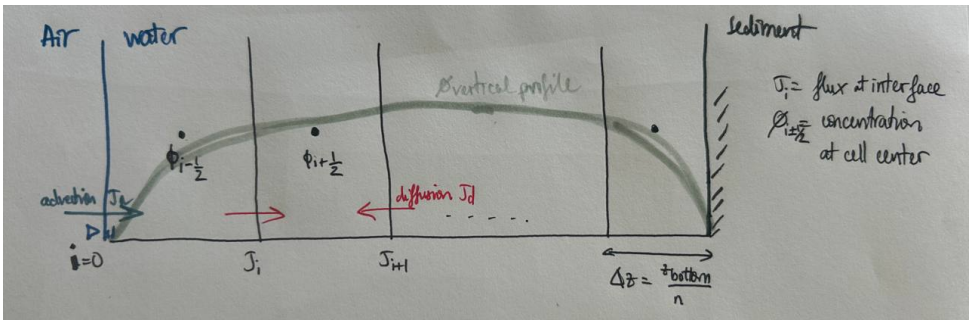


Figure 2: Vertical discretization of the model.

The concentration of state variable $\phi_{i\pm\frac{1}{2}}$ or hereafter referred as ϕ_i is defined at the cell center of each cell. The fluxes J_i consisting of advection $J_{a,i}$ and diffusion $J_{d,i}$ flux are defined at the interfaces between cells.

The equations for advective and diffusive flux in interior cells are below.

$$J_i = w\,\frac{\phi_i + \phi_{i+1}}{2} - \kappa_i \frac{\phi_{i+1} - \phi_i}{\Delta z} \tag{12}$$

Using a finite-volume formulation, the discretized equation for ϕ is blelow.

$$\frac{d\phi_i}{dt} = -\frac{J_i - J_{i-1}}{\Delta z}$$

Boundary cells are treated using no-flux boundary conditions at the surface and bottom. This means that the advection and diffusion flux at surface and bottom i.e., $J_{a,0}$, $J_{a,n}$, $J_{d,0}$, $J_{d,n}$, are assumed to be 0, except for O_2.

Table 2: Boundary conditions for state variables.

Variable	**Surface flux (J_0)**	**Bottom flux (J_n)**
Nutrients (N)	=0	=0
Phytoplankton (P)	=0	=0
Zooplankton (Z)	=0	=0
Detritus (D)	=0	=0
Oxygen (O)	$= k_{O_2,\text{surf}}(O_{2,\text{eq}} - O_0)$	$= -k_{O_2,\text{bottom}}(O_{\text{sed}} - O_n)$

5 Solution convergence

To check for solution convergence and results robustness, the model is first run without seasonality introduced and constant light IO = 100 μmol ph/m^2/sec. The system contains a uniform concentration of 1 mmolN/m^3 for all state variables N, P, Z, D across the vertical profile.

Figure 3 shows the solution without seasonality converges with relatively no change in the concentration for 5 state variables after approximately 50 days.

The result robustness is check by looking at how changing the vertical discretization affects the modelled results. Figure 4 shows that there is virtually difference between the 2 solutions with n=10 or n=1000, where the coarse resolution results shows a more segmented or discretized profile, especially before the solution converges. This shows that the model in a long run simulation is robust when using n=100.

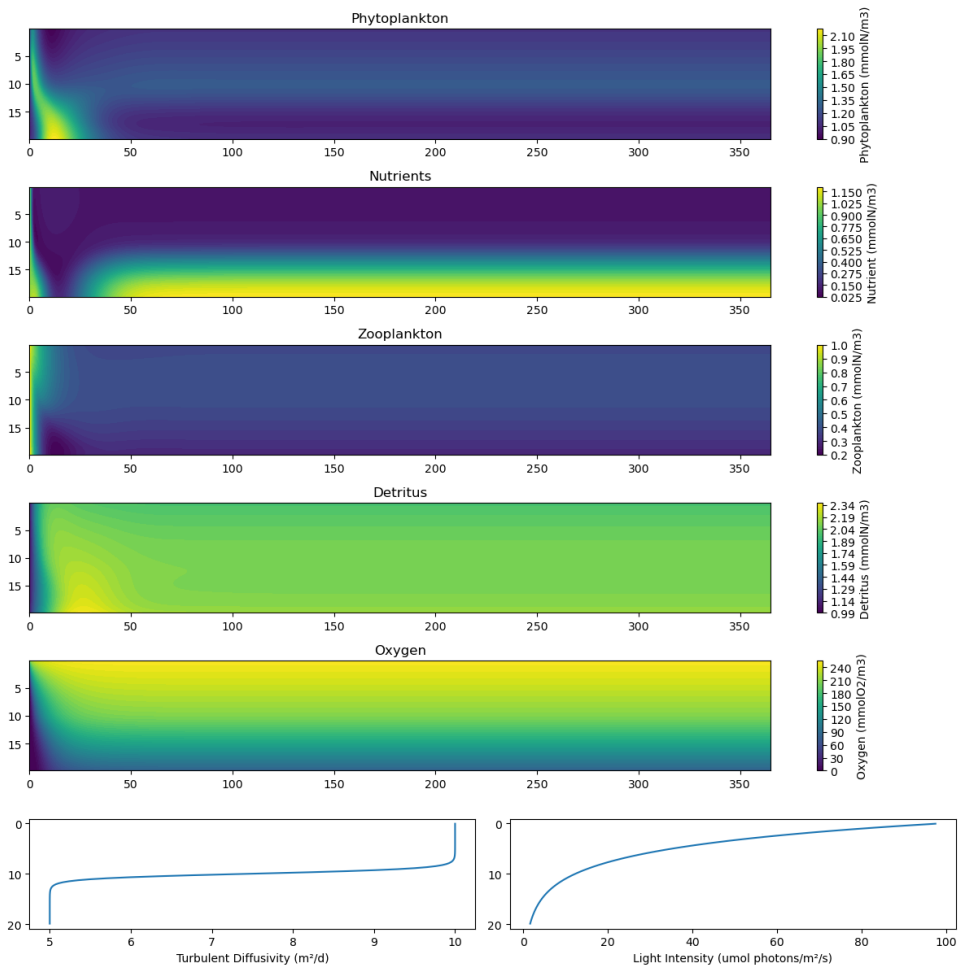
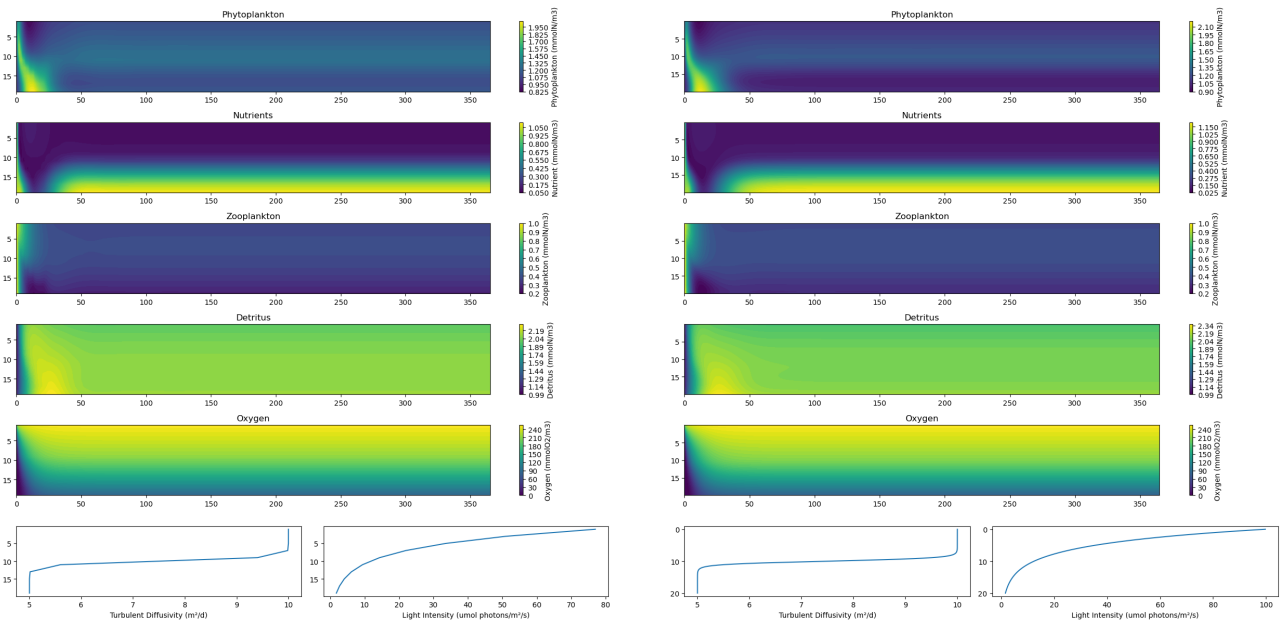


Figure 3: Solution convergence check, IO = 100 μmol ph/m^2/sec, $N_0 = P_0 = Z_0 = D_0 = 1$ mmolN/m^3.

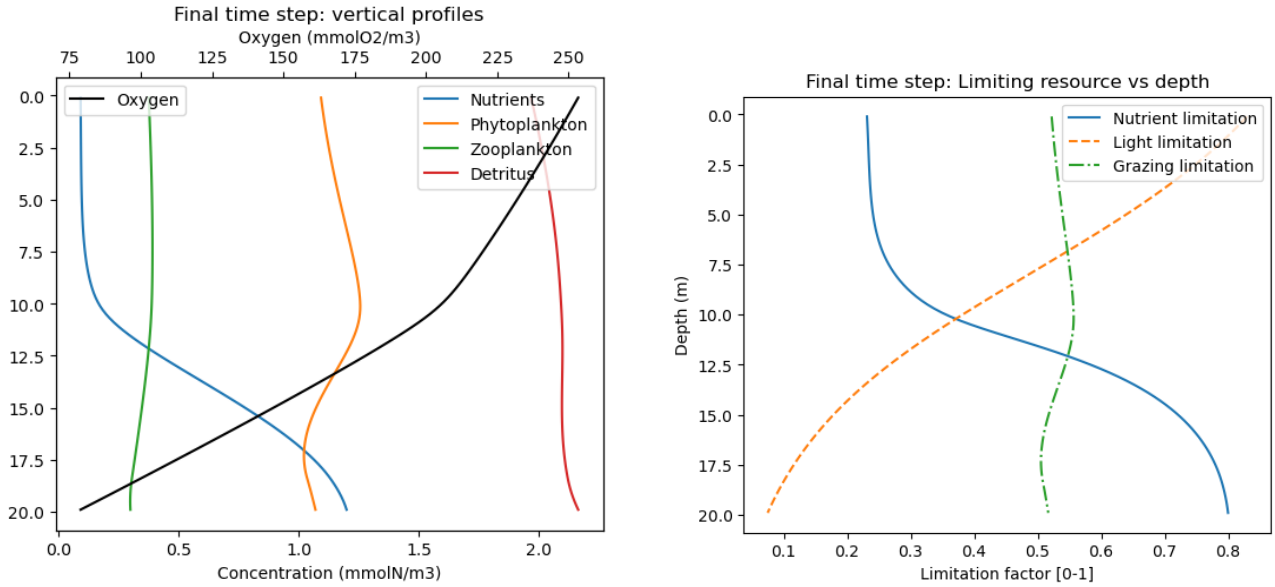


(a) n=10 (b) n=1000

Figure 4: Result robustness check, IO = 100 μmol ph/m^2/sec, $N_0 = P_0 = Z_0 = D_0 = 1$ mmolN/m^3.

6 Limiting factor: phytoplankton growth

Figure 5 shows the concentration of state variables and limiting factor at the final time step. Phytoplankton growth is dependent on nutrient and light. The intersection of nutrient and light limiting factor indicates the depth at which peak phytoplankton concentration occurs. Figure 5b shows the intersection at depth of approximately 10 m which coincides with the peak concentration of phytoplankton shown in Figure 5a.



(a) State variables

(b) Limiting factors

Figure 5: Concentration at final simulation time, IO = 100 μmol ph/m^2/sec, $N_0 = P_0 = Z_0 = D_0 =$ 1 mmolN/m^3.

7 Steady state solutions in average summer

Using the same model setup as the convergence check, except that nutrient concentration was initialized with $N_0 = 10$ mmolN/m^3, the modelled results are shown in Figure 6 for the vertical profile over time. The vertical profile over time shows an oscillatory solution which is likely due to the predator-prey relationship between phytoplankton and zooplankton where low phytoplankton concentration coincides with high concentration of zooplankton and vice versa. Nutrients are in abundance, as such, the detritus pool is in abundance. Along with the rate of mineralization of detritus allowing for enough nutrients to be re-introduced in the water system for phytoplankton uptake, phytoplankton continues to grow and be eaten by zooplankton. This is confirmed by checking on model sensitivity to re-mineralization rate $\tau = 0.01$ day^{-1} shown in Figure 9, discussed later in Section 8.2.

Regardless, Figure 7 shows final simulation time with $N_0 = 10$ mmolN/m^3. Compare with Figure 5, phytoplankton concentration is higher (at approx. 4 mmolN/m^3) with higher nutrient concentration. As such, there is a upper shift in the light limiting function where there is less light reaching bottom water likely due to self-shading by phytoplankton. With higher nutrient loadings, Figure 7b shows the intersection between light and nutrient limiting function for phytoplankton growth at water depth of <5 m which coincides with the peak concentration for phytoplankton at similar depth in Figure 7a. Additionally, high nutrient loading also drives higher zooplankton and detritus concentration as well. This is due to higher mass of phytoplankton leads to higher mass of zooplankton, and subsequently higher detritus due to mortality and egestion. As there is more phytoplankton near surface, there appears to be a less drastic decline in oxygen concentration at the top 5 m depth comparing Figure 5a to Figure 7a which is likely due to oxygen yielded from phytoplankton growth.

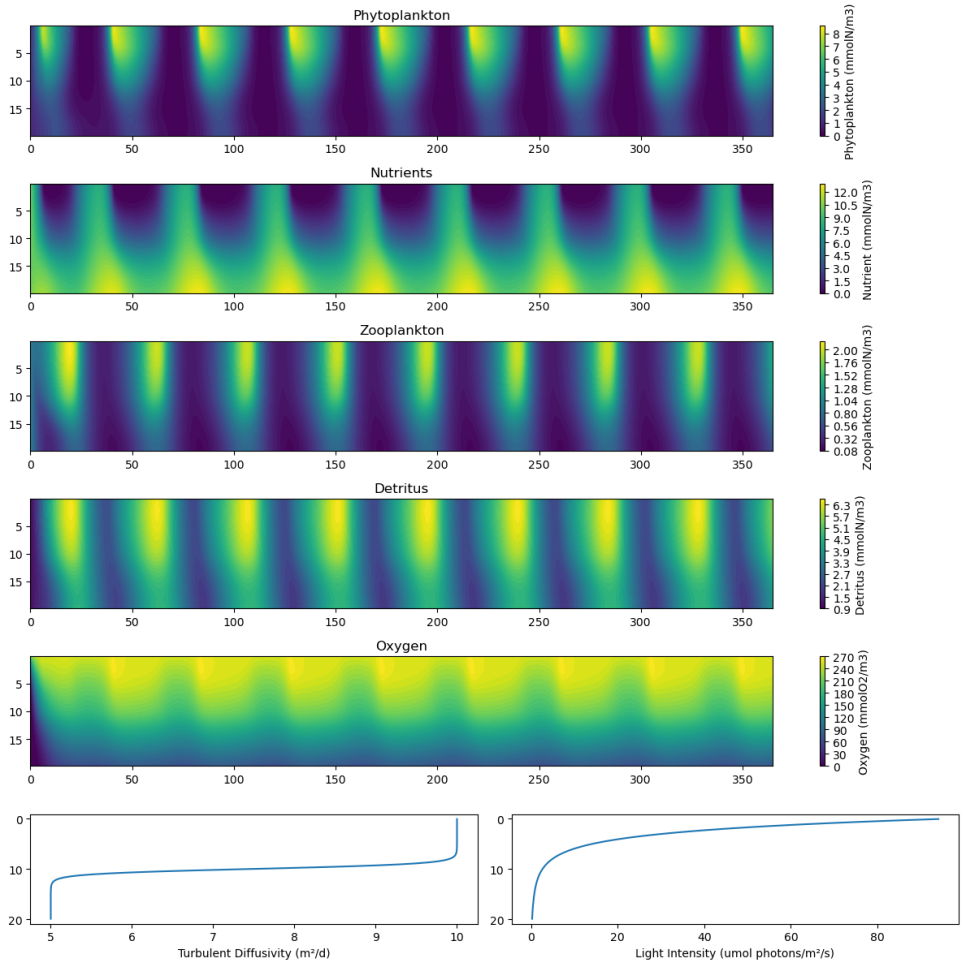
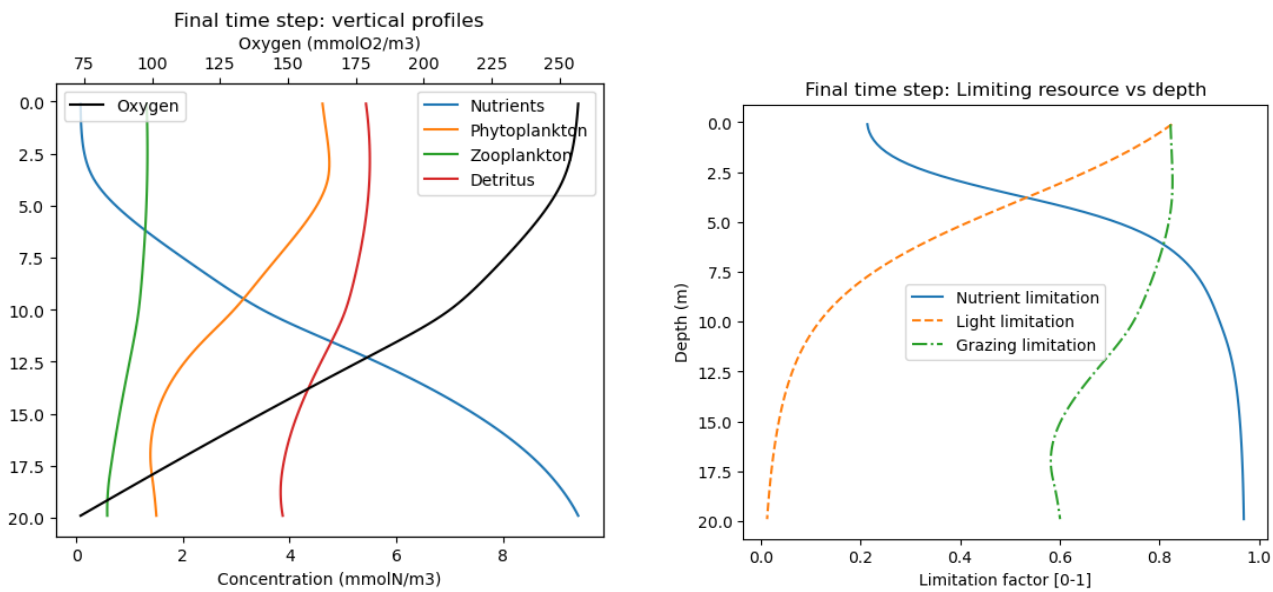


Figure 6: IO = 100 μmol ph/m^2/sec, $P_0 = Z_0 = D_0 = 1$ mmolN/m^3, $N_0 = 10$ mmolN/m^3.



(a) State variables

(b) Limiting factors

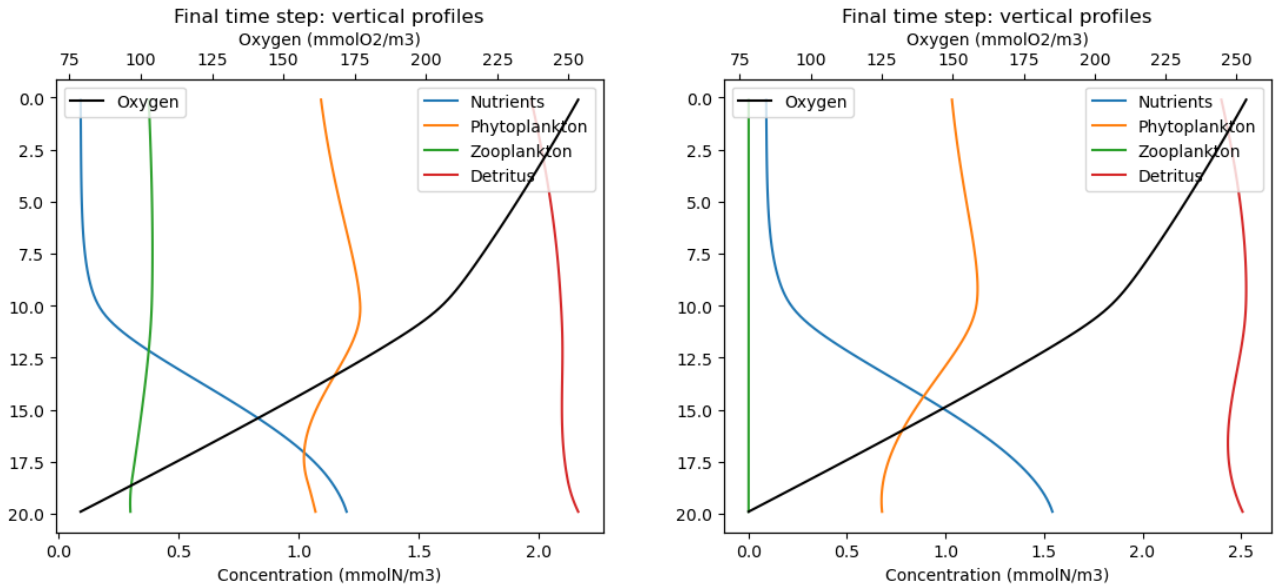
Figure 7: Concentration at final simulation time, IO = 100 μmol ph/m^2/sec, $P_0 = Z_0 = D_0 = 1$ mmolN/m^3, $N_0 = 10$ mmolN/m^3.

8 Sensitivity analysis

The sensitivity for the rate of detritus remineralization to nitrogen is described and explained in Section 7. Additionally, the sensitivity for additional parameters are shown here.

8.1 Phytoplankton mortality $m_{\mathbf{P}}$

Figure 8 shows the steady state concentration for simulation with m_{P}=0.07 day^{-1} (Figure 8a similar to Figure 5a) and with m_{P}=0.25 day^{-1} (Figure 8b). Note the difference in the x-axis scale for state variables concentration mmolN/m^3. The vertical distribution of state variables remain the same with peak phytoplankton concentration remains at approx. 10 m depth. However, with higher mortality rate m_{P}=0.25 day^{-1}, there is lower concentration of phytoplankton overall with concentration at bottom <1 m_{P}=0.25 day^{-1}. As there is less phytoplankton, zooplankton is relatively non-existent in the water system. In contrast, due to higher phytoplankton mortality, there is higher detritus concentration as well as higher nutrient concentration due to lower phytoplankton uptake.



(a) m_{P}=0.07 day^{-1}

(b) m_{P}=0.25 day^{-1}

Figure 8: IO = 100 μmol ph/m^2/sec, $\mathrm{N_0} = \mathrm{P_0} = \mathrm{Z_0} = \mathrm{D_0} = 1$ mmolN/m^3.

8.2 Remineralization rate τ

As discussed in 7, Figure 9 shows the system with high nutrient loading $\mathrm{N_0} = 10$ mmolN/m^3 but lower remineralization rate $\tau = 0.01$ day^{-1}. The figure shows that the water column reaches a steady state where concentration no longer changes over time. As there is lower mineralization rate, nutrient is not re-introduced into the system fast enough for phytoplankton to uptake. As such, phytoplankton concentration does not increase again where zooplankton can graze more (as seen in the oscillatory solution in Figure 6). A reduced remineralization rate essentially decouples nutrient recycling from biological production, leading to detritus accumulation, lower long-term productivity, and reduced oxygen consumption.

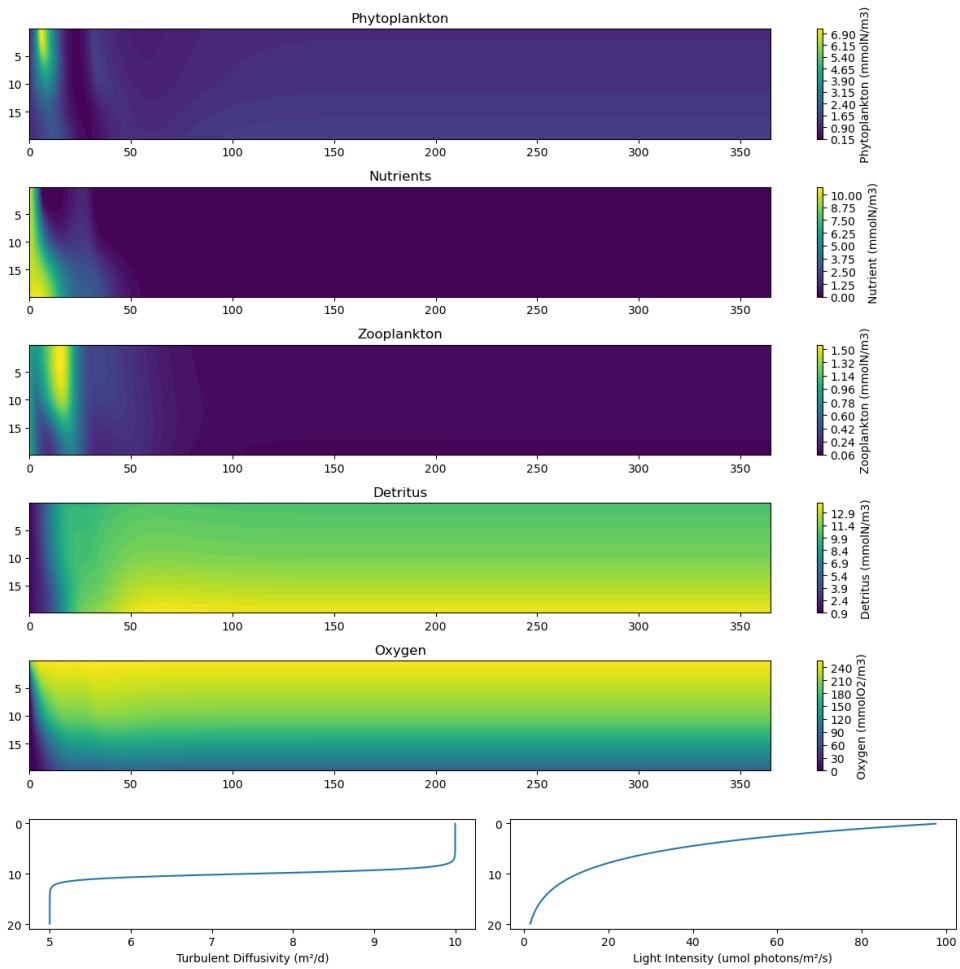


Figure 9: IO = 100 μmol ph/m^2/sec, $P_0 = Z_0 = D_0 = 1$ mmolN/m^3, $N_0 = 10$ mmolN/m^3, $\tau = 0.01$ day^{-1}.

9 Seasonality

Both seasonality in light and diffusivity is included in this section to realistically describe the stratification event in the water column due to light and subsequent mixing in the water column. The simulation includes seasonality in surface light with peak summer incident light $I_{0,\max} = 1400$ μmol ph/m^2/secFigure 10.

The figure shows a distinct seasonal pattern despite only changes in seasonal light. During wintertime where there is low light availability, there is relatively low existence of phytoplankton as such limited zooplankton and detritus. As there is minimal phytoplankton growth during this period due to light limiting, nutrient availability in the water is high. Additionally, due to low phytoplankton concentration, oxygen concentration also reduces at near surface. When springtime comes with increasing light, there observes a phytoplankton bloom around the surface top 5 m in the water column. Due to phytoplankton growth, nitrogen uptake is high hence low nitrogen concentration in the water column. A period after the spring bloom (approx. 100 days) where phytoplankton concentration stablizes, there is an increase in detritus concentration at the bottom, likely due to the phytoplankton mortality as well as zooplankton egestion as zooplankton grazes on phytoplankton. This detritus pool is subsequently re-mineralized, contribution to nutrients slowly re-introduced into the water column.

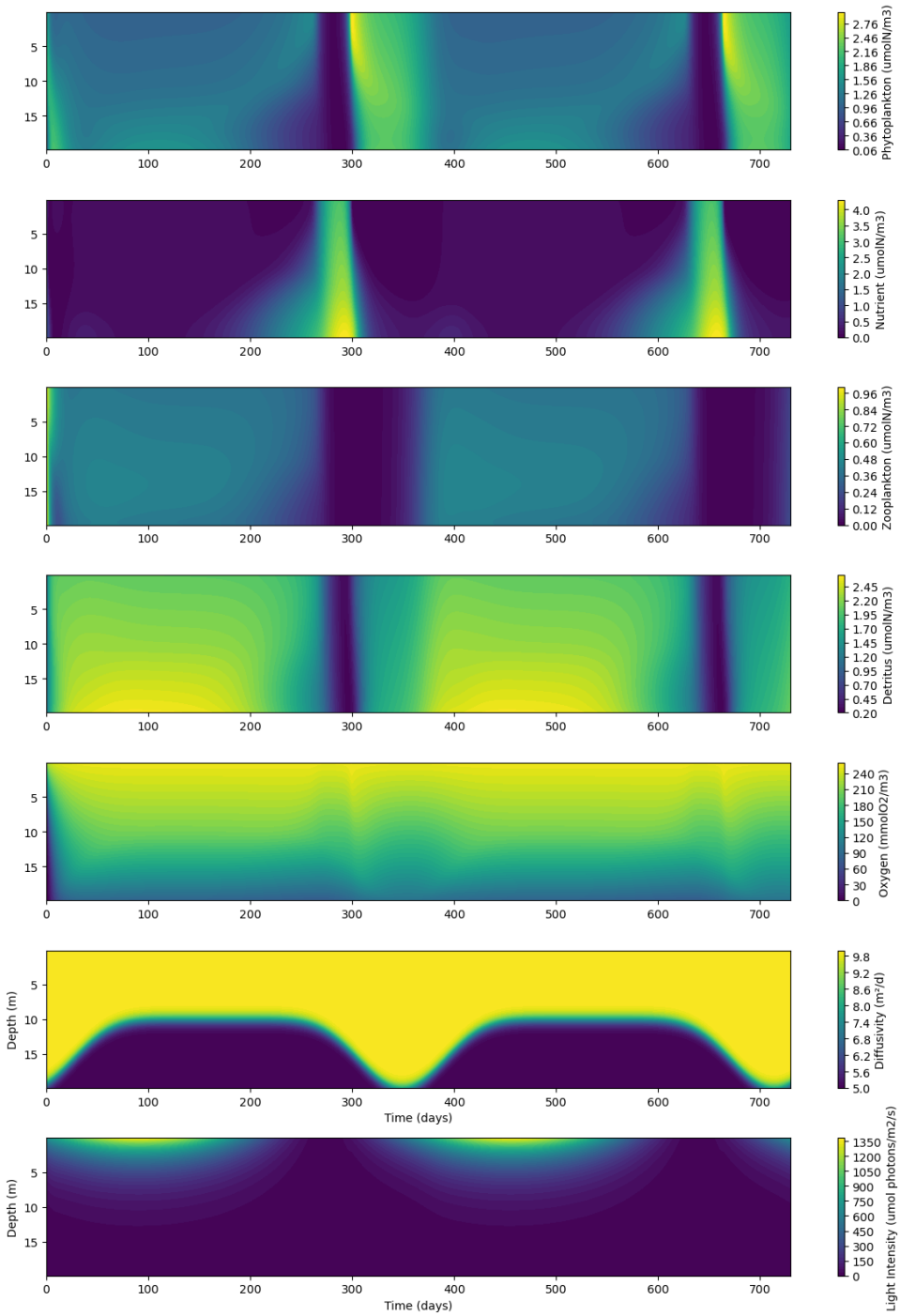


Figure 10: Seasonality in surface light and diffusivity

10 Model improvement

The model is currently a closed system, i.e. imagining the water bed is similar to concrete with no exchange at bottom. As such, the concentration of state variables generally accumulate at bottom water overtime, unless mixed by seasonality in light and diffusivity. In order to improve the model, a change of bottom fluxes for state variable should be revisited to account for the diffusion from nutrient-rich sediment to the water column. Furthermore, additional processes related to sediment, e.g. aerobic, anaerobic processes, etc., can be included as a separate sediment model to interact with the water column model. To make the model even more realistic, the mixing layer can be dynamically modelled using additional climatic forcings of temperature and salinity.

11 GitHub

Link to the GitHub repository for this report can be found here: `https://github.com/baongocthai/25314-Computational-Marine-Ecological-Modelling-Spring-2026.git`